

Functions: prototype, call, return, types ★ Golden

One college period · 40–45 minutes · Programming using C

What you will learn

- State advantages and characteristics of functions
- Differentiate built-in and user-defined; call by value vs reference
- Write programs that use functions (greater, factorial, area)

Function. A named, reusable block that performs a task. `main()` is itself a function. Advantages: modularity, reuse, easier testing, team work.

Prototype (declaration) → Definition (body) → Call (use)

```
int square(int n);           /* prototype */
void main() {
    printf("%d", square(5)); /* call – actual parameter 5 */
}
int square(int n) {          /* definition – n is formal parameter */
    return n * n;
}
```

Idea	Meaning
Built-in	<code>printf</code> , <code>scanf</code> , <code>sqrt</code> , <code>strlen</code> — already in libraries
User-defined	You write them
Return type	<code>void</code> if nothing is returned
Formal parameters	In the definition header
Actual parameters	In the call
Call by value	A copy is passed; original safe
Call by reference	Address (&) / pointer; function can change the original
Local vs global	Local inside block; global outside functions
Recursion	Function calls itself (factorial, Fibonacci)

HOW BIEK ASKS THIS

2026 long Q5: main characteristics of a function with example. List: name, return type, parameters, independent compilation, reuse, call by value/reference.

اردو میں سمجھیں

فنکشن نام والا بلاک ہے جو بار بار کام کرتا ہے۔ Prototype اعلان، تعریف جسم، کال استعمال۔ `return` قدر بھیجتا ہے۔

CHECK YOURSELF

1. Does main have to exist?
2. sqrt comes from which header?

Answers: 1. Yes — it is the entry point · 2. math.h

MCQs

1. A subprogram that returns a value is a:
 - a. function
 - b. report
 - c. macro only
 - d. file
2. Passing a copy of a variable is:
 - a. pass by value
 - b. pass by address
 - c. pass by reference
 - d. pass by pointer

Key: 1=a, 2=a

Short questions (3 marks each)

1. Write any one program using a function to print the greater of two numbers.
2. Differentiate formal and actual parameters.
3. What are built-in functions? Give three.

Model short answers

1. `int greater(int x,int y){ return (x>y)?x:y; }` call from main after `scanf`.
2. Formal: names in the definition. Actual: values in the call. They correspond in order and type.
3. Library functions: `printf`, `scanf`, `getchar`, `strlen`, `sqrt`.

Long questions (board Section C)

1. What are the main characteristics of a function? Describe with an example program.